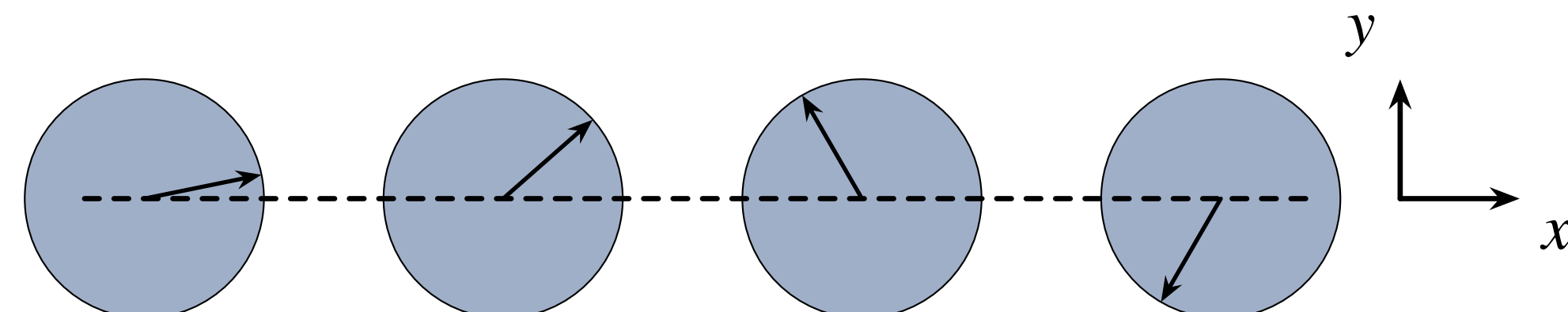


Premise

We use the time-independent Schrödinger Equation to extract eigenvalues and eigenvectors, with hamiltonian:

$$\hat{H} = B\hat{K} + g\hat{V}, \quad B, g : \text{adjustable parameters} \quad (1)$$

Where the hamiltonian describes a coplanar rotor system.



We solve Schrödinger equation by diagonalizing the hamiltonian over the product basis of single site plane waves:

$$\hat{H} |\psi\rangle = E |\psi\rangle \quad (2)$$

However, this hamiltonian become extremely large really fast which makes exact diagonalization computationally heavy. (Even though we are currently working with systems of rotors the method is equal applicable to magnetic systems.)

Problem Statement

Generalize Coupled Cluster methods (CC) to rotor systems, to explore larger systems. Compare with other approaches and also consider Thermal Properties.

Improving Exact Diagonalization

In order to explore larger systems with Exact Diagonalization and Configuration Coupled Cluster (CCC) we use Natural Orbitals:

- Truncate the basis using one particle reduced density matrix for small system (2-3 sites), and use selected eigenvectors (Natural Orbital, NO).
- Use sparse matrix techniques to construct Hamiltonian and use direct diagonalization.
- Systems up to about 9 sites with 5 NO's can be studied.

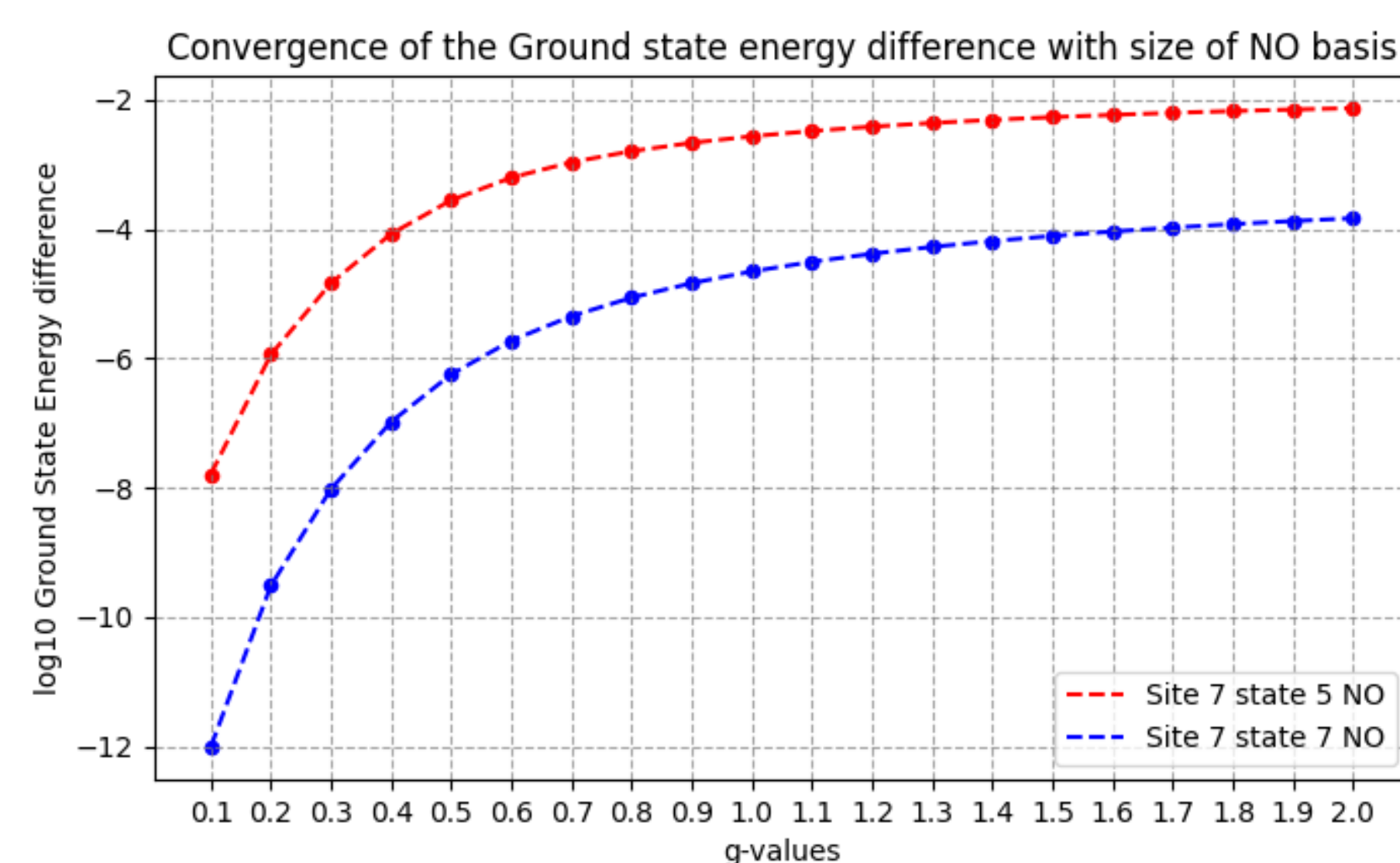


Figure 1: Accuracy of the NO with 5 and 7 NO basis truncation for $g = 0.2$.

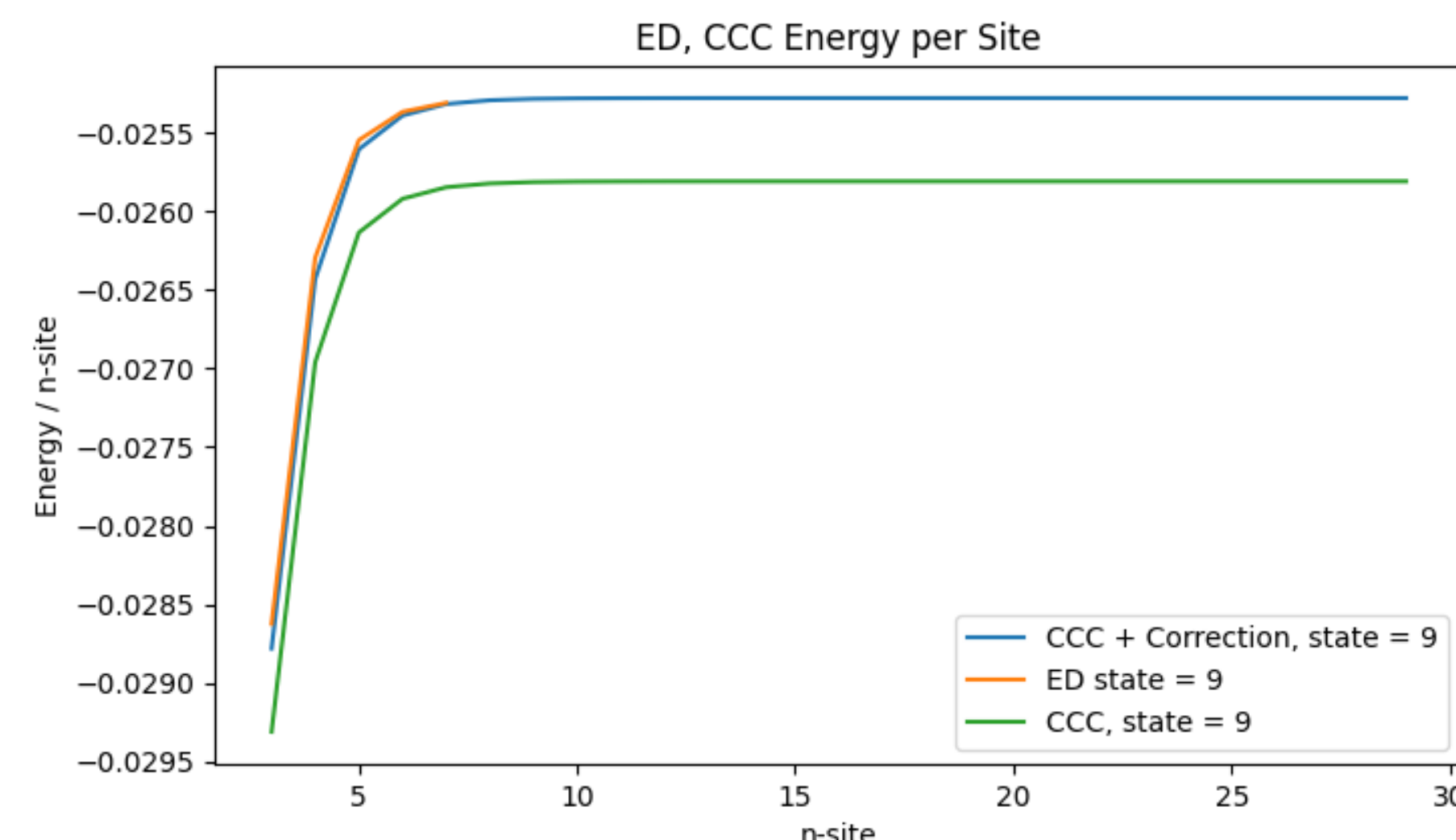


Figure 2: Accuracy CCC as n-site increases $g = 0.2$.

Periodic vs. Non-Periodic System Setup

CCC calculations can be used for both periodic and finite systems. One could go beyond nearest neighbor interactions. CCC calculations for periodic systems are more efficient and converge faster with the number of sites.

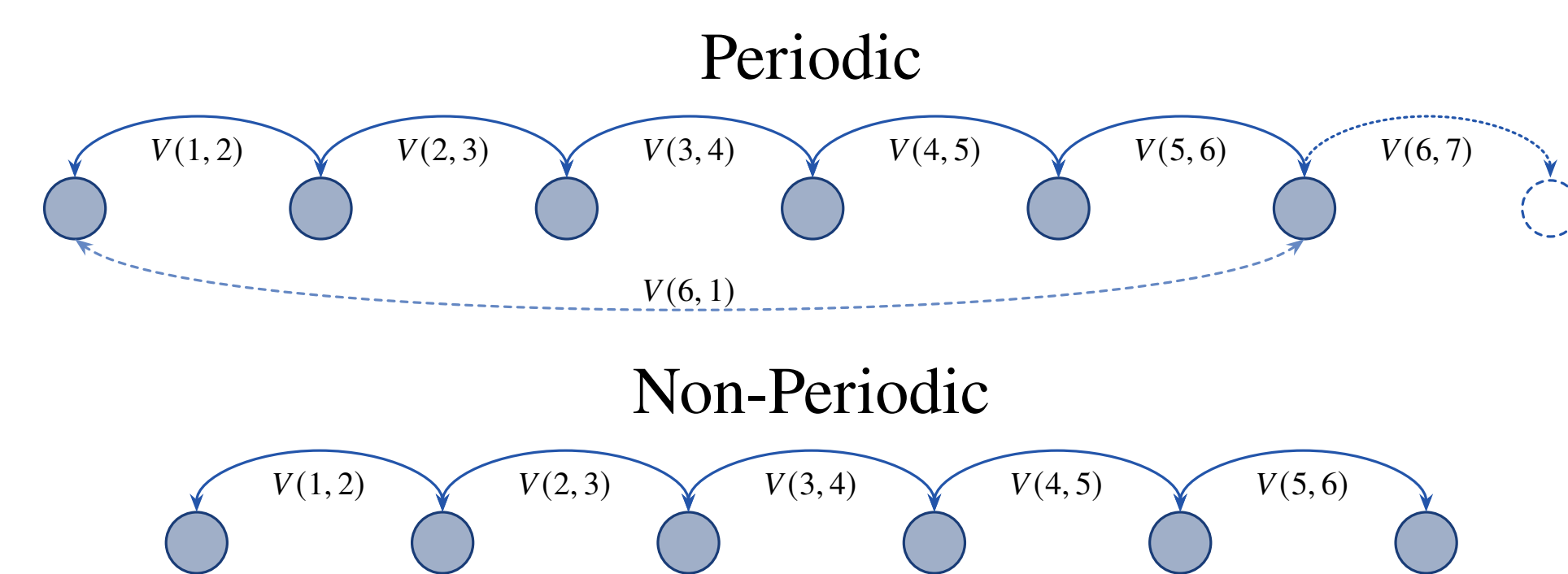


Figure 3: Circles represent rotor sites, arrows represent site interactions between sites 1 and 2; 2 and 3; etc.

Configurational Coupled Cluster

We define our $\hat{T}_1$ and $\hat{T}_2$ operators in terms of single-site excitation operators $\hat{e}_i^a$ where i indicates a single low-lying state:

$$\hat{T}_1 = \sum_x \sum_{a,i} t_i^a(x) \hat{e}_i^a(x) \quad (3)$$

$$\hat{T}_2 = \sum_{x,y} \sum_{a,b,i,j} t_{ij}^{ab}(x,y) \hat{e}_i^a(x) \hat{e}_j^b(y) \quad (4)$$

The $\hat{T}_1$ and $\hat{T}_2$ operators form the overall $\hat{T}$ in the CC ansatz:

$$|\psi\rangle = e^{\hat{T}} |0\rangle \quad (5)$$

Substituting this form into the Schrödinger Equation and projecting against the basis set $|\tau\rangle$, the equations for the cluster amplitudes can be expressed as:

$$\langle \tau | e^{-\hat{T}} \hat{H} e^{\hat{T}} | 0 \rangle = E \delta_{\tau 0} \quad (6)$$

By expanding one can arrive at a full set of algebraic equations which are solved numerically through iterative and/or imaginary time propagation procedures.

Features of CCC:

- method is size extensive
- the number of parameters scales like $n_{state}^2 * n_{site}$ for periodic systems
- method is only accurate up to $g \approx 0.4$

Computational Efficiency Results

To optimize the algorithm of the CCC solution, all tensor operators are defined as matrix operators to make use of BLAS procedure:

Table 1: Computation time comparison for periodic vs non-periodic systems. ($g = 0.2$).

(a) Non-Periodic

System states – 5	
# Sites	Time per iter./site (s)
11	2.12 sec 10^{-1}
21	1.19 sec
System sites – 5	
# States	Time per iter./site (s)
11	8.19 sec 10^{-2}
15	2.43 sec 10^{-1}

(b) Dense matrices, BLAS

System states – 5	
# Sites	Time per iter./site (s)
11	8.06 sec 10^{-2}
21	4.85 sec 10^{-1}
System sites – 5	
# States	Time per iter./site (s)
11	1.56 sec 10^{-2}
15	3.15 sec 10^{-2}

Table 2: CCC Computation time comparison for periodic system. ($g = 0.2$).

(a) Sparse matrices		(b) Dense matrices, BLAS	
System states – 11		System states – 11	
# Sites	Time per iter./site	# Sites	Time per iter./site (s)
10	7.22 sec 10^{-2}	10	2.32 (sec 10^{-3})
100	2.61 sec	100	9.21 (sec 10^{-4})
System sites – 3		System sites – 3	
# States	Time per iter./site	# States	Time per iter./site (s)
11	1.74 sec 10^{-2}	11	8.95 (sec 10^{-4})
151	5.85 sec	151	11.4 (sec)

Variation of $0 < g < 1$

Range $0 < g < 1$ we use a delocalized reference state of $|0\rangle$ and get accurate results for $g < 0.4$.

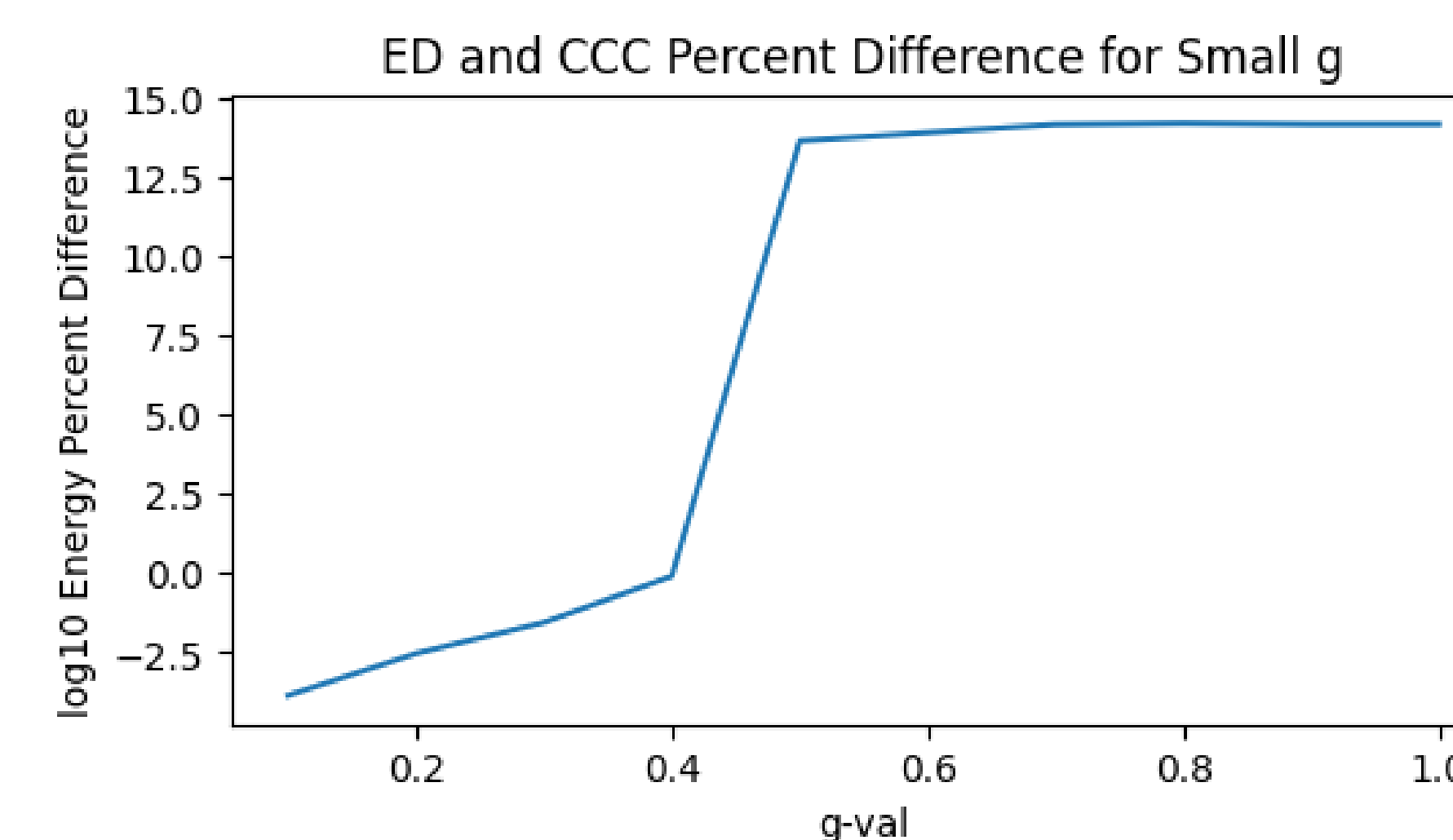


Figure 4: As g increases the CCC results loose accuracy.

Very Large $g \geq 100$ (or Small B) Limit

To explore the very large g we can use $B < 0.01$ instead. The system becomes a superposition of two states.

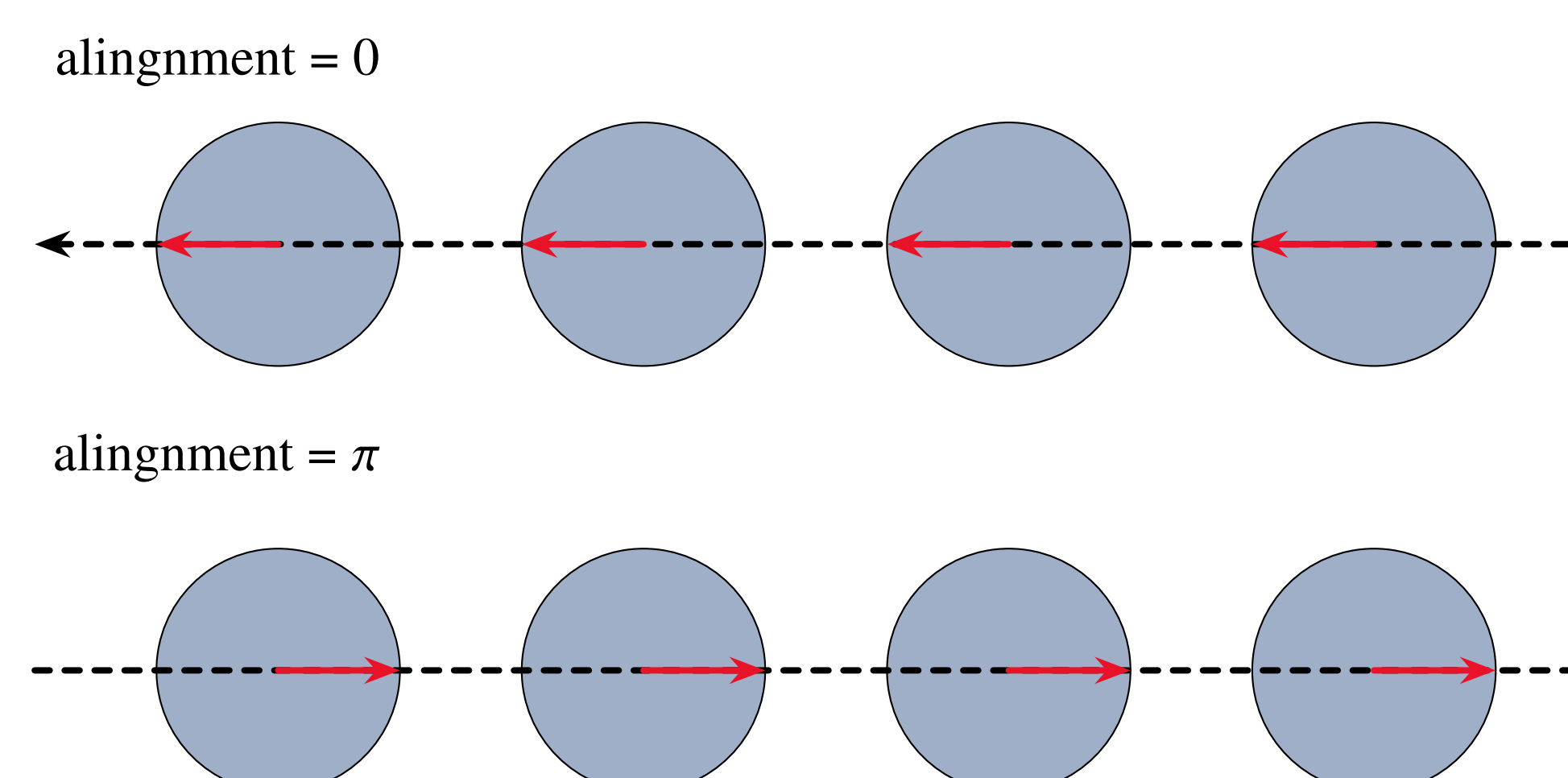


Figure 5: The schematic for superposition of the two state alignments.

There are two degenerate states that are both delocalized. They can alternatively be localized and one of these states (after obtaining the Natural Orbitals) is then used as a reference state in a broken symmetry CCC calculation.

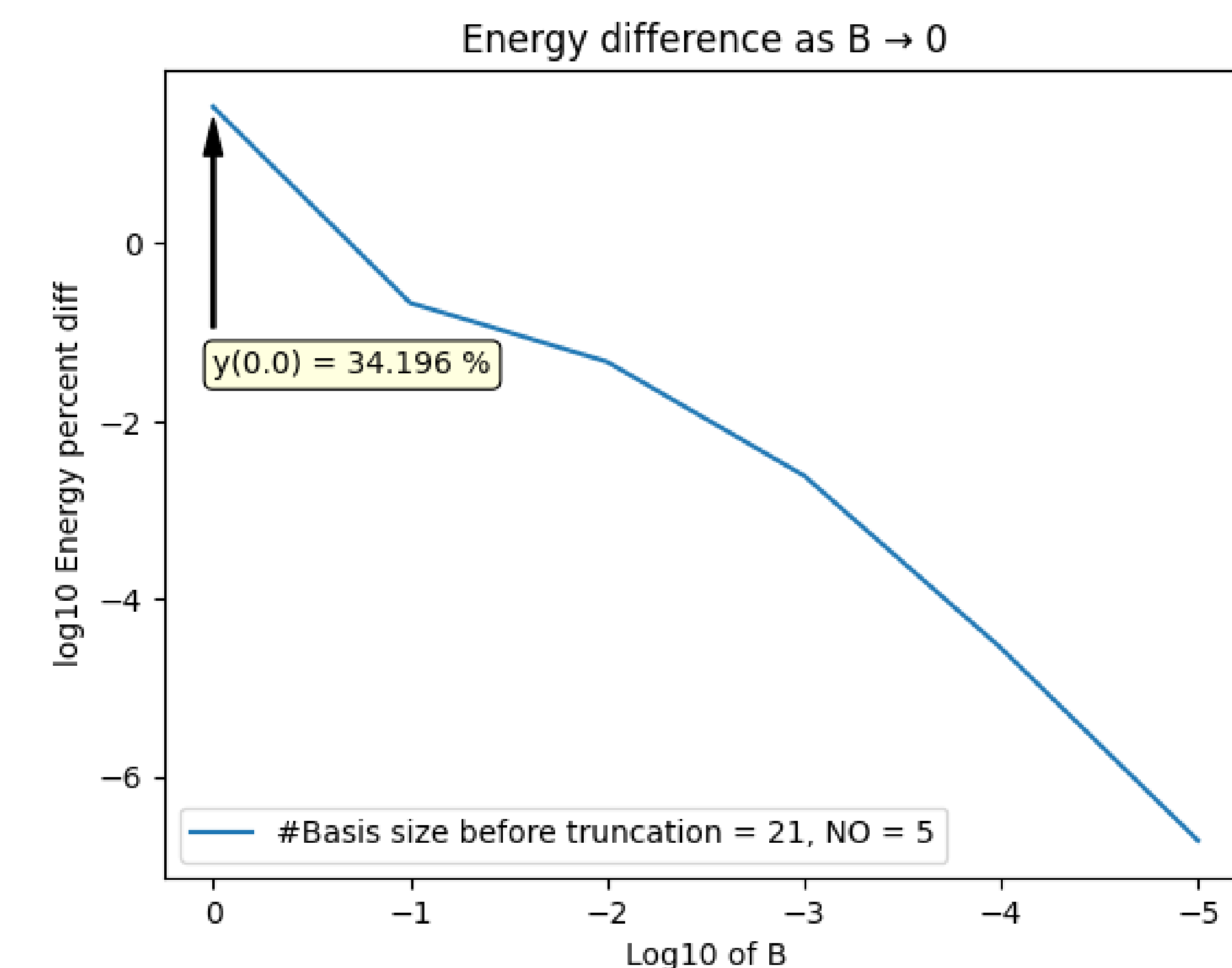
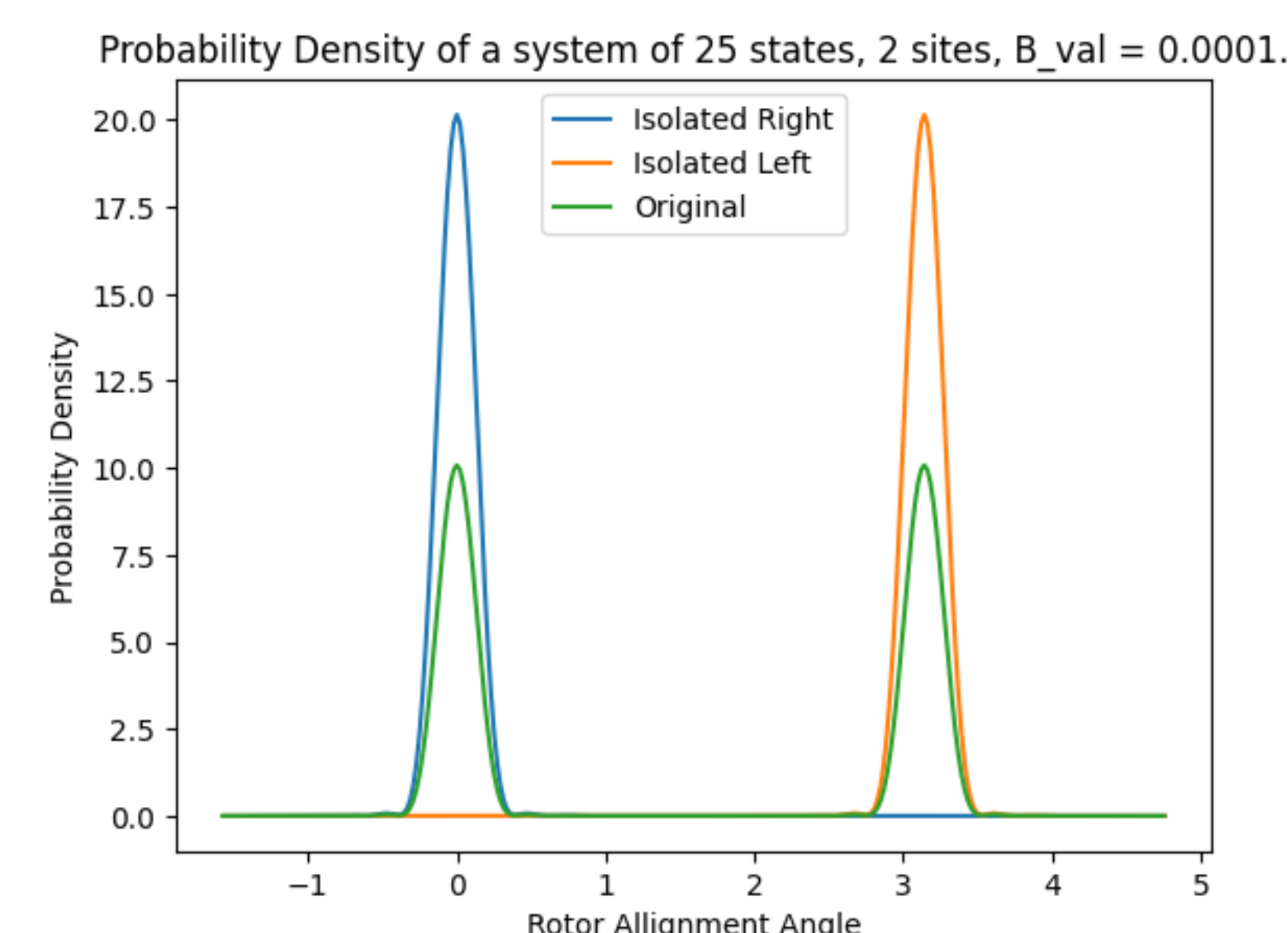


Figure 6: As B approaches 0 (g approaches infinity) accuracy increases.

Future Work

- Exploration of additional rotor systems:
 - Twisted planar rotors
 - Water molecule clusters
- Extension to Magnetic Systems (1-dimensional)
- Investigation of thermal properties
- Study of systems with up to approximately 9 sites and 3–5 natural orbitals (NOs)
- Extension to two/three-dimensional lattices

References

- [1] Moed, M. Shaeer; Brown, James; Ibrahim, Alexander; Vilas Boas De Oliveira, Estevao; Roy, Pierre-Nicholas. (2025). *Qubit encodings for lattices of dipolar planar rotors*. arXiv preprint arXiv:2507.17952 [quant-ph].
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